# ВЕЖБЕ 4

1. Случајна променљива $Z$ има нормалну $N(0,1)$ расподелу. Израчунати вероватноће

- $P(Z < 1{,}645) = \Phi(1{,}645) =$

## 4.1 Gausova normalna raspodela $\mathcal{N}(0,1)$

$$\phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

| $z$ | .00 | .01 | .02 | .03 | .04 | .05 | .06 | .07 | .08 | .09 |
|---|---|---|---|---|---|---|---|---|---|---|
| .0 | .5000 | .5040 | .5080 | .5120 | .5160 | .5199 | .5239 | .5279 | .5319 | .5359 |
| .1 | .5398 | .5438 | .5478 | .5517 | .5557 | .5596 | .5636 | .5675 | .5714 | .5753 |
| .2 | .5793 | .5832 | .5871 | .5910 | .5948 | .5987 | .6026 | .6064 | .6103 | .6141 |
| .3 | .6179 | .6217 | .6255 | .6293 | .6331 | .6368 | .6406 | .6443 | .6480 | .6517 |
| .4 | .6554 | .6591 | .6628 | .6664 | .6700 | .6736 | .6772 | .6808 | .6844 | .6879 |
| .5 | .6915 | .6950 | .6985 | .7019 | .7054 | .7088 | .7123 | .7157 | .7190 | .7224 |
| .6 | .7257 | .7291 | .7324 | .7357 | .7389 | .7422 | .7454 | .7486 | .7517 | .7549 |
| .7 | .7580 | .7611 | .7642 | .7673 | .7703 | .7734 | .7764 | .7794 | .7823 | .7852 |
| .8 | .7881 | .7910 | .7939 | .7967 | .7995 | .8023 | .8051 | .8079 | .8106 | .8133 |
| .9 | .8159 | .8186 | .8212 | .8238 | .8264 | .8289 | .8315 | .8340 | .8365 | .8389 |
| 1.0 | .8413 | .8438 | .8461 | .8485 | .8508 | .8531 | .8554 | .8577 | .8599 | .8621 |
| 1.1 | .8643 | .8665 | .8686 | .8708 | .8729 | .8749 | .8770 | .8790 | .8810 | .8830 |
| 1.2 | .8849 | .8869 | .8888 | .8907 | .8925 | .8944 | .8962 | .8980 | .8997 | .9015 |
| 1.3 | .9032 | .9049 | .9066 | .9082 | .9099 | .9115 | .9131 | .9147 | .9162 | .9177 |
| 1.4 | .9192 | .9207 | .9222 | .9237 | .9251 | .9265 | .9279 | .9292 | .9306 | .9319 |
| 1.5 | .9332 | .9345 | .9357 | .9370 | .9382 | .9394 | .9406 | .9418 | .9429 | .9441 |
| 1.6 | .9452 | .9463 | .9474 | .9485 | .9495 | .9505 | .9515 | .9525 | .9535 | .9545 |
| 1.7 | .9554 | .9564 | .9573 | .9582 | .9591 | .9599 | .9608 | .9616 | .9625 | .9633 |
| 1.8 | .9641 | .9649 | .9656 | .9664 | .9671 | .9678 | .9686 | .9693 | .9699 | .9706 |
| 1.9 | .9713 | .9719 | .9726 | .9732 | .9738 | .9744 | .9750 | .9756 | .9762 | .9767 |
| 2.0 | .9773 | .9778 | .9783 | .9788 | .9793 | .9798 | .9803 | .9808 | .9812 | .9817 |

1. Случајна променљива $Z$ има нормалну $N(0,1)$ расподелу. Израчунати вероватноће

- $P(Z<1{,}645)=\Phi(1{,}645)=\frac{0{,}9495+0{,}9505}{2}=0{,}95$

1. Случајна променљива $Z$ има нормалну $N(0,1)$ расподелу. Израчунати вероватноће

- $P(Z<1{,}645)=\Phi(1{,}645)=\frac{0{,}9495+0{,}9505}{2}=0{,}95$

Приметимо да су у таблици на дну стране изузете вредности квантила $z_\alpha$ (заокружене на три децимале) за најчешће коришћене вероватноће: $0.9, 0.95, 0.975, 0.99, \ldots$

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 1.4 | .9192 | .9207 | .9222 | .9237 | .9251 | .9265 | .9279 | .9292 | .9306 | .9319 |
| 1.5 | .9332 | .9345 | .9357 | .9370 | .9382 | .9394 | .9406 | .9418 | .9429 | .9441 |
| 1.6 | .9452 | .9463 | .9474 | .9485 | .9495 | .9505 | .9515 | .9525 | .9535 | .9545 |
| 1.7 | .9554 | .9564 | .9573 | .9582 | .9591 | .9599 | .9608 | .9616 | .9625 | .9633 |
| 1.8 | .9641 | .9649 | .9656 | .9664 | .9671 | .9678 | .9686 | .9693 | .9699 | .9706 |
| 1.9 | .9713 | .9719 | .9726 | .9732 | .9738 | .9744 | .9750 | .9756 | .9762 | .9767 |
| 2.0 | .9773 | .9778 | .9783 | .9788 | .9793 | .9798 | .9803 | .9808 | .9812 | .9817 |
| 2.1 | .9821 | .9826 | .9830 | .9834 | .9838 | .9842 | .9846 | .9850 | .9854 | .9857 |
| 2.2 | .9861 | .9864 | .9868 | .9871 | .9875 | .9878 | .9881 | .9884 | .9887 | .9890 |
| 2.3 | .9893 | .9896 | .9898 | .9901 | .9904 | .9906 | .9909 | .9911 | .9913 | .9916 |
| 2.4 | .9918 | .9920 | .9922 | .9925 | .9927 | .9929 | .9931 | .9932 | .9934 | .9936 |
| 2.5 | .9938 | .9940 | .9941 | .9943 | .9945 | .9946 | .9948 | .9949 | .9951 | .9952 |
| 2.6 | .9953 | .9955 | .9956 | .9957 | .9959 | .9960 | .9961 | .9962 | .9963 | .9964 |
| 2.7 | .9965 | .9966 | .9967 | .9968 | .9969 | .9970 | .9971 | .9972 | .9973 | .9974 |
| 2.8 | .9974 | .9975 | .9976 | .9977 | .9977 | .9978 | .9979 | .9980 | .9980 | .9981 |
| 2.9 | .9981 | .9982 | .9983 | .9983 | .9984 | .9984 | .9985 | .9985 | .9986 | .9986 |
| 3.0 | .9987 | .9987 | .9987 | .9988 | .9988 | .9989 | .9989 | .9989 | .9990 | .9990 |
| 3.1 | .9990 | .9991 | .9991 | .9991 | .9992 | .9992 | .9992 | .9992 | .9993 | .9993 |
| 3.2 | .9993 | .9993 | .9994 | .9994 | .9994 | .9994 | .9994 | .9995 | .9995 | .9995 |
| 3.3 | .9995 | .9995 | .9996 | .9996 | .9996 | .9996 | .9996 | .9996 | .9996 | .9997 |
| 3.4 | .9997 | .9997 | .9997 | .9997 | .9997 | .9997 | .9997 | .9997 | .9998 | .9998 |

| $z$ | 1.282 | 1.645 | 1.960 | 2.326 | 2.576 | 3.090 | 3.291 | 3.891 | 4.417 |
|---|---|---|---|---|---|---|---|---|---|
| $\phi(z)$ | .9 | .95 | .975 | .99 | .995 | .999 | .9995 | .99995 | .999995 |
| $2(1-\phi(z))$ | .20 | .10 | .05 | .02 | .01 | .002 | .001 | .0001 | .00001 |

1. Случајна променљива $Z$ има нормалну $N(0,1)$ расподелу. Израчунати вероватноће

- $P(Z<1{,}645)=\Phi(1{,}645)=\frac{0{,}9495+0{,}9505}{2}=0{,}95$

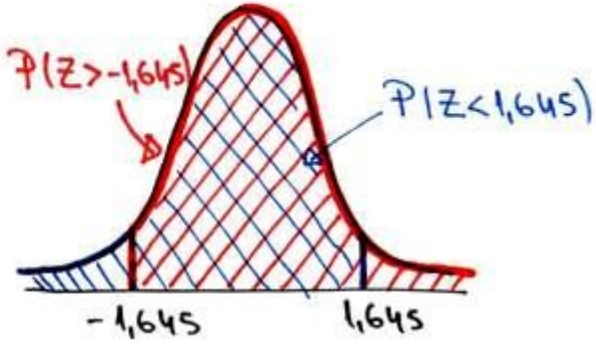
Приметимо да су у табличи на дну стране изузетне вредности квантила $z$ (заокружене на три децимале) за најчешће коришћене вероватноће: $0.9, 0.95, 0.975, 0.99, \ldots$

- $P(Z>-1{,}645)=1-P(Z\le -1{,}645)=1-\Phi(-1{,}645)=1-(1-\Phi(1{,}645))=\Phi(1{,}645)=0{,}95$ $\quad \Phi(-z)=1-\Phi(z)$

- $P(|Z|<1{,}645)=P(-1{,}645<Z<1{,}645)=\Phi(1{,}645)-\Phi(-1{,}645)=$
$\Phi(1{,}645)-(1-\Phi(1{,}645))=2\Phi(1{,}645)-1=2\cdot 0{,}95-1=0{,}9$

$P(|Z|<z)=2\Phi(z)-1$

- $P(|Z|>1{,}96)=1-P(|Z|\le 1{,}96)=1-2\Phi(1{,}96)+1=2-2\Phi(1{,}96)=$
$2-2\cdot 0{,}975=0{,}05$



## 4.1 Gausova normalna raspodela $\mathcal{N}(0,1)$

$$\phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

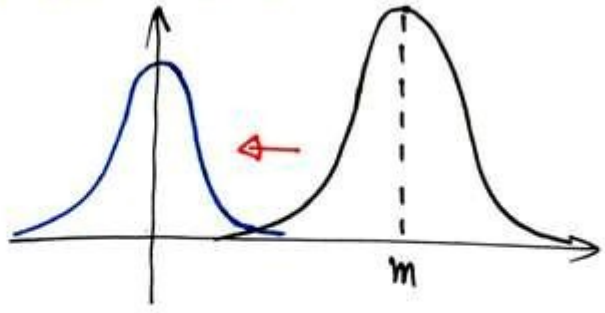
| $z$ | .00 | .01 | .02 | .03 | .04 | .05 | .06 | .07 | .08 | .09 |
|---|---|---|---|---|---|---|---|---|---|---|
| .0 | .5000 | .5040 | .5080 | .5120 | .5160 | .5199 | .5239 | .5279 | .5319 | .5359 |
| .1 | .5398 | .5438 | .5478 | .5517 | .5557 | .5596 | .5636 | .5675 | .5714 | .5753 |
| .2 | .5793 | .5832 | .5871 | .5910 | .5948 | .5987 | .6026 | .6064 | .6103 | .6141 |
| .3 | .6179 | .6217 | .6255 | .6293 | .6331 | .6368 | .6406 | .6443 | .6480 | .6517 |
| .4 | .6554 | .6591 | .6628 | .6664 | .6700 | .6736 | .6772 | .6808 | .6844 | .6879 |
| .5 | .6915 | .6950 | .6985 | .7019 | .7054 | .7088 | .7123 | .7157 | .7190 | .7224 |
| .6 | .7257 | .7291 | .7324 | .7357 | .7389 | .7422 | .7454 | .7486 | .7517 | .7549 |
| .7 | .7580 | .7611 | .7642 | .7673 | .7703 | .7734 | .7764 | .7794 | .7823 | .7852 |
| .8 | .7881 | .7910 | .7939 | .7967 | .7995 | .8023 | .8051 | .8079 | .8106 | .8133 |
| .9 | .8159 | .8186 | .8212 | .8238 | .8264 | .8289 | .8315 | .8340 | .8365 | .8389 |
| 1.0 | .8413 | .8438 | .8461 | .8485 | .8508 | .8531 | .8554 | .8577 | .8599 | .8621 |
| 1.1 | .8643 | .8665 | .8686 | .8708 | .8729 | .8749 | .8770 | .8790 | .8810 | .8830 |
| 1.2 | .8849 | .8869 | .8888 | .8907 | .8925 | .8944 | .8962 | .8980 | .8997 | .9015 |
| 1.3 | .9032 | .9049 | .9066 | .9082 | .9099 | .9115 | .9131 | .9147 | .9162 | .9177 |
| 1.4 | .9192 | .9207 | .9222 | .9237 | .9251 | .9265 | .9279 | .9292 | .9306 | .9319 |
| 1.5 | .9332 | .9345 | .9357 | .9370 | .9382 | .9394 | .9406 | .9418 | .9429 | .9441 |
| 1.6 | .9452 | .9463 | .9474 | .9485 | .9495 | .9505 | .9515 | .9525 | .9535 | .9545 |
| 1.7 | .9554 | .9564 | .9573 | .9582 | .9591 | .9599 | .9608 | .9616 | .9625 | .9633 |
| 1.8 | .9641 | .9649 | .9656 | .9664 | .9671 | .9678 | .9686 | .9693 | .9699 | .9706 |
| 1.9 | .9713 | .9719 | .9726 | .9732 | .9738 | .9744 | .9750 | .9756 | .9762 | .9767 |
| 2.0 | .9773 | .9778 | .9783 | .9788 | .9793 | .9798 | .9803 | .9808 | .9812 | .9817 |
| 2.1 | .9821 | .9826 | .9830 | .9834 | .9838 | .9842 | .9846 | .9850 | .9854 | .9857 |
| 2.2 | .9861 | .9864 | .9868 | .9871 | .9875 | .9878 | .9881 | .9884 | .9887 | .9890 |
| 2.3 | .9893 | .9896 | .9898 | .9901 | .9904 | .9906 | .9909 | .9911 | .9913 | .9916 |
| 2.4 | .9918 | .9920 | .9922 | .9925 | .9927 | .9929 | .9931 | .9932 | .9934 | .9936 |
| 2.5 | .9938 | .9940 | .9941 | .9943 | .9945 | .9946 | .9948 | .9949 | .9951 | .9952 |
| 2.6 | .9953 | .9955 | .9956 | .9957 | .9959 | .9960 | .9961 | .9962 | .9963 | .9964 |
| 2.7 | .9965 | .9966 | .9967 | .9968 | .9969 | .9970 | .9971 | .9972 | .9973 | .9974 |
| 2.8 | .9974 | .9975 | .9976 | .9977 | .9977 | .9978 | .9979 | .9980 | .9980 | .9981 |
| 2.9 | .9981 | .9982 | .9983 | .9983 | .9984 | .9984 | .9985 | .9985 | .9986 | .9986 |
| 3.0 | .9987 | .9987 | .9987 | .9988 | .9988 | .9989 | .9989 | .9989 | .9990 | .9990 |
| 3.1 | .9990 | .9991 | .9991 | .9991 | .9992 | .9992 | .9992 | .9992 | .9993 | .9993 |
| 3.2 | .9993 | .9993 | .9994 | .9994 | .9994 | .9994 | .9994 | .9995 | .9995 | .9995 |
| 3.3 | .9995 | .9995 | .9996 | .9996 | .9996 | .9996 | .9996 | .9996 | .9996 | .9997 |
| 3.4 | .9997 | .9997 | .9997 | .9997 | .9997 | .9997 | .9997 | .9997 | .9998 | .9998 |

| $z$ | 1.282 | 1.645 | 1.960 | 2.326 | 2.576 | 3.090 | 3.291 | 3.891 | 4.417 |
|---|---|---|---|---|---|---|---|---|---|
| $\phi(z)$ | .9 | .95 | .975 | .99 | .995 | .999 | .9995 | .99995 | .999995 |
| $2(1-\phi(z))$ | .20 | .10 | .05 | .02 | .01 | .002 | .001 | .0001 | .00001 |

2. Случајна променљива $X$ има нормалну $\mathcal{N}(1.5, 2.5)$ расподелу. Израчунати вероватноће $P(X<1,645)$, $P(X>-1,645)$, $P(|X|<1,645)$, $P(|X|>1,96)$

$X: \mathcal{N}(1.5, 2.5)$ $\quad m = 1.5$, $\sigma = 2.5$

$$X^* = \frac{X-m}{\sigma} = \frac{X-1,5}{2,5} : \mathcal{N}(0,1)$$




- $P(X<1,645) = P\left(X^* < \frac{1,645-1,5}{2,5}\right) = P(X^* < 0,058) \approx 0,5239$ $\quad$ R: 0,5231
- $P(X>-1,645) = 1 - P(X \leq -1,645) = 1 - P(X^* \leq -1,258) = P(X^* \leq 1,258) \approx 0,8962$ $\quad$ R: 0,8958
- $P(|X|<1,645) = P(-1,645 < X < 1,645) = P(-1,258 < X^* < 0,058) = \Phi(0,058) - \Phi(-1,258) = 0,4201$
- $P(|X|>1,96) = 1 - P(|X| \leq 1,96) = 1 - P(-1,96 < X < 1,96) = 1 - P(-1,384 < X^* < 0,184) =$

$$1 - \Phi(0,184) + \Phi(-1,384) = 2 - \Phi(0,184) - \Phi(1,384) = 2 - 0,5714 - 0,9162 = 0,5124$$

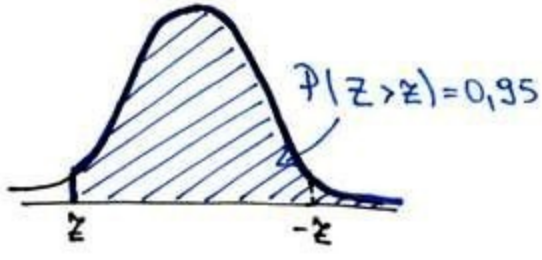
3. Случајна променљива $Z$ има нормалну $N(0,1)$ расподелу. Наћи вредности $z$ тако да је

- $P(Z < z) = 0,9$

$z$ је између 1,28 и 1,29 → друга табличка $z = 1,282$

- $P(Z > z) = 0,95$

$0,95 = P(Z > z) = 1 - P(Z \leq z) \Rightarrow P(Z \leq z) = 0,05$



| $z$ | .00 | .01 | .02 | .03 | .04 | .05 | .06 | .07 | .08 | .09 |
|---|---|---|---|---|---|---|---|---|---|---|
| .0 | .5000 | .5040 | .5080 | .5120 | .5160 | .5199 | .5239 | .5279 | .5319 | .5359 |
| .1 | .5398 | .5438 | .5478 | .5517 | .5557 | .5596 | .5636 | .5675 | .5714 | .5753 |
| .2 | .5793 | .5832 | .5871 | .5910 | .5948 | .5987 | .6026 | .6064 | .6103 | .6141 |
| .3 | .6179 | .6217 | .6255 | .6293 | .6331 | .6368 | .6406 | .6443 | .6480 | .6517 |
| .4 | .6554 | .6591 | .6628 | .6664 | .6700 | .6736 | .6772 | .6808 | .6844 | .6879 |
| .5 | .6915 | .6950 | .6985 | .7019 | .7054 | .7088 | .7123 | .7157 | .7190 | .7224 |
| .6 | .7257 | .7291 | .7324 | .7357 | .7389 | .7422 | .7454 | .7486 | .7517 | .7549 |
| .7 | .7580 | .7611 | .7642 | .7673 | .7703 | .7734 | .7764 | .7794 | .7823 | .7852 |
| .8 | .7881 | .7910 | .7939 | .7967 | .7995 | .8023 | .8051 | .8079 | .8106 | .8133 |
| .9 | .8159 | .8186 | .8212 | .8238 | .8264 | .8289 | .8315 | .8340 | .8365 | .8389 |
| 1.0 | .8413 | .8438 | .8461 | .8485 | .8508 | .8531 | .8554 | .8577 | .8599 | .8621 |
| 1.1 | .8643 | .8665 | .8686 | .8708 | .8729 | .8749 | .8770 | .8790 | .8810 | .8830 |
| 1.2 | .8849 | .8869 | .8888 | .8907 | .8925 | .8944 | .8962 | .8980 | .8997 | .9015 |
| 1.3 | .9032 | .9049 | .9066 | .9082 | .9099 | .9115 | .9131 | .9147 | .9162 | .9177 |
| 1.4 | .9192 | .9207 | .9222 | .9237 | .9251 | .9265 | .9279 | .9292 | .9306 | .9319 |
| 1.5 | .9332 | .9345 | .9357 | .9370 | .9382 | .9394 | .9406 | .9418 | .9429 | .9441 |
| 1.6 | .9452 | .9463 | .9474 | .9485 | .9495 | .9505 | .9515 | .9525 | .9535 | .9545 |
| 1.7 | .9554 | .9564 | .9573 | .9582 | .9591 | .9599 | .9608 | .9616 | .9625 | .9633 |
| 1.8 | .9641 | .9649 | .9656 | .9664 | .9671 | .9678 | .9686 | .9693 | .9699 | .9706 |
| 1.9 | .9713 | .9719 | .9726 | .9732 | .9738 | .9744 | .9750 | .9756 | .9762 | .9767 |
| 2.0 | .9773 | .9778 | .9783 | .9788 | .9793 | .9798 | .9803 | .9808 | .9812 | .9817 |
| 2.1 | .9821 | .9826 | .9830 | .9834 | .9838 | .9842 | .9846 | .9850 | .9854 | .9857 |
| 2.2 | .9861 | .9864 | .9868 | .9871 | .9875 | .9878 | .9881 | .9884 | .9887 | .9890 |
| 2.3 | .9893 | .9896 | .9898 | .9901 | .9904 | .9906 | .9909 | .9911 | .9913 | .9916 |
| 2.4 | .9918 | .9920 | .9922 | .9925 | .9927 | .9929 | .9931 | .9932 | .9934 | .9936 |
| 2.5 | .9938 | .9940 | .9941 | .9943 | .9945 | .9946 | .9948 | .9949 | .9951 | .9952 |
| 2.6 | .9953 | .9955 | .9956 | .9957 | .9959 | .9960 | .9961 | .9962 | .9963 | .9964 |
| 2.7 | .9965 | .9966 | .9967 | .9968 | .9969 | .9970 | .9971 | .9972 | .9973 | .9974 |
| 2.8 | .9974 | .9975 | .9976 | .9977 | .9977 | .9978 | .9979 | .9980 | .9980 | .9981 |
| 2.9 | .9981 | .9982 | .9983 | .9983 | .9984 | .9984 | .9985 | .9985 | .9986 | .9986 |
| 3.0 | .9987 | .9987 | .9987 | .9988 | .9988 | .9989 | .9989 | .9989 | .9990 | .9990 |
| 3.1 | .9990 | .9991 | .9991 | .9991 | .9992 | .9992 | .9992 | .9992 | .9993 | .9993 |
| 3.2 | .9993 | .9993 | .9994 | .9994 | .9994 | .9994 | .9994 | .9995 | .9995 | .9995 |
| 3.3 | .9995 | .9995 | .9996 | .9996 | .9996 | .9996 | .9996 | .9996 | .9996 | .9997 |
| 3.4 | .9997 | .9997 | .9997 | .9997 | .9997 | .9997 | .9997 | .9997 | .9998 | .9998 |

| $z$ | 1.282 | 1.645 | 1.960 | 2.326 | 2.576 | 3.090 | 3.291 | 3.891 | 4.417 |
|---|---|---|---|---|---|---|---|---|---|
| $\phi(z)$ | .9 | .95 | .975 | .99 | .995 | .999 | .9995 | .99995 | .999995 |
| $2(1-\phi(z))$ | .20 | .10 | .05 | .02 | .01 | .002 | .001 | .0001 | .00001 |

3. Случајна променљива $Z$ има нормалну $N(0,1)$ расподелу. Наћи вредности $z$ тако да је

- $P(Z < z) = 0{,}9$

$z$ је између 1,28 и 1,29 → друга табличка $z = 1{,}282$

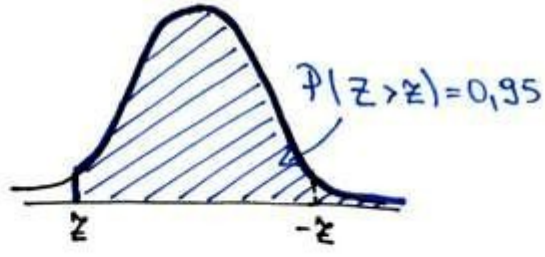
- $P(Z > z) = 0{,}95$

$0{,}95 = P(Z > z) = 1 - P(Z \leq z) \Rightarrow P(Z \leq z) = 0{,}05$

→ $z$ је негативно

Сада је $P(Z \leq -z) = 0{,}95$

$-z = 1{,}645$, па је $z = -1{,}645$




- $P(|Z| < z) = 0{,}95$

$0{,}95 = P(|Z| < z) = 2\Phi(z) - 1$

$\Phi(z) = 0{,}975 \Rightarrow z = 1{,}96$

4. Случајна променљива $X$ има нормалну $\mathcal{N}(1.5, 2.5)$ расподелу. Наћи вредност $x$ тако да је $P(X<x)=0,9$ и $P(X>x)=0,95$.

$$X^* = \frac{X-1,5}{2,5} : \mathcal{N}(0,1)$$

- $P(X<x)=0,9$

$$P(X<x) = P\left(X^* < \frac{x-1,5}{2,5}\right) = 0,9$$

$\Rightarrow$

## 4.1 Gausova normalna raspodela $\mathcal{N}(0,1)$

$$\phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

| $z$ | .00 | .01 | .02 | .03 | .04 | .05 | .06 | .07 | .08 | .09 |
|---|---|---|---|---|---|---|---|---|---|---|
| .0 | .5000 | .5040 | .5080 | .5120 | .5160 | .5199 | .5239 | .5279 | .5319 | .5359 |
| .1 | .5398 | .5438 | .5478 | .5517 | .5557 | .5596 | .5636 | .5675 | .5714 | .5753 |
| .2 | .5793 | .5832 | .5871 | .5910 | .5948 | .5987 | .6026 | .6064 | .6103 | .6141 |
| .3 | .6179 | .6217 | .6255 | .6293 | .6331 | .6368 | .6406 | .6443 | .6480 | .6517 |
| .4 | .6554 | .6591 | .6628 | .6664 | .6700 | .6736 | .6772 | .6808 | .6844 | .6879 |
| .5 | .6915 | .6950 | .6985 | .7019 | .7054 | .7088 | .7123 | .7157 | .7190 | .7224 |
| .6 | .7257 | .7291 | .7324 | .7357 | .7389 | .7422 | .7454 | .7486 | .7517 | .7549 |
| .7 | .7580 | .7611 | .7642 | .7673 | .7703 | .7734 | .7764 | .7794 | .7823 | .7852 |
| .8 | .7881 | .7910 | .7939 | .7967 | .7995 | .8023 | .8051 | .8079 | .8106 | .8133 |
| .9 | .8159 | .8186 | .8212 | .8238 | .8264 | .8289 | .8315 | .8340 | .8365 | .8389 |
| 1.0 | .8413 | .8438 | .8461 | .8485 | .8508 | .8531 | .8554 | .8577 | .8599 | .8621 |
| 1.1 | .8643 | .8665 | .8686 | .8708 | .8729 | .8749 | .8770 | .8790 | .8810 | .8830 |
| 1.2 | .8849 | .8869 | .8888 | .8907 | .8925 | .8944 | .8962 | .8980 | .8997 | .9015 |
| 1.3 | .9032 | .9049 | .9066 | .9082 | .9099 | .9115 | .9131 | .9147 | .9162 | .9177 |
| 1.4 | .9192 | .9207 | .9222 | .9237 | .9251 | .9265 | .9279 | .9292 | .9306 | .9319 |
| 1.5 | .9332 | .9345 | .9357 | .9370 | .9382 | .9394 | .9406 | .9418 | .9429 | .9441 |
| 1.6 | .9452 | .9463 | .9474 | .9485 | .9495 | .9505 | .9515 | .9525 | .9535 | .9545 |
| 1.7 | .9554 | .9564 | .9573 | .9582 | .9591 | .9599 | .9608 | .9616 | .9625 | .9633 |
| 1.8 | .9641 | .9649 | .9656 | .9664 | .9671 | .9678 | .9686 | .9693 | .9699 | .9706 |
| 1.9 | .9713 | .9719 | .9726 | .9732 | .9738 | .9744 | .9750 | .9756 | .9762 | .9767 |
| 2.0 | .9773 | .9778 | .9783 | .9788 | .9793 | .9798 | .9803 | .9808 | .9812 | .9817 |

4. Случајна променљива $X$ има нормалну $\mathcal{N}(1.5, 2.5)$ расподелу. Наћи вредност $x$ тако да је $P(X<x)=0,9$ и $P(X>x)=0,95$.

$$X^* = \frac{X-1,5}{2,5} : \mathcal{N}(0,1)$$

- $P(X<x)=0,9$

$$P(X<x) = P\left(X^* < \frac{x-1,5}{2,5}\right) = 0,9$$

$$\Rightarrow 1,282 = \frac{x-1,5}{2,5}$$

$$x = 4,705$$

- $P(X>x)=0,95$

$$P(X>x) = P\left(X^* > \frac{x-1,5}{2,5}\right) = 0,95$$

$$\Rightarrow \frac{x-1,5}{2,5} = -1,645$$

$$x = -2,6125$$

5. Случајна променљива $Z$ има нормалну $\mathcal{N}(0,1)$ расподелу и нека је $X = Z^2$. Израчунати вероватноћу $P(X < 2{,}25)$ и наћи вредност $x$ за коју је $P(X > x) = 0{,}95$.

$Z_1, Z_2, \ldots, Z_n$ независне случајне променљиве, $Z_i : \mathcal{N}(0,1)$, $i = 1,2,\ldots,n$

ако је $X = \sum_{i=1}^{n} Z_i^2$, онда $X$ има Пирсонову $\chi^2$ расподелу, $X : \chi^2_n$ — $n$-број степени слободе

$\Rightarrow X : \chi^2_1$

$P(X < 2{,}25) \approx 0{,}9$

## 4.3 Pirsonova $\chi^2$ raspodela $F(y) = \int_0^y (x^{\frac{n}{2}-1} e^{-\frac{x}{2}})/(2^{\frac{n}{2}}\Gamma(n/2))dx$

| $n$ \ $F$ | .0050 | .0100 | .0250 | .0500 | .1000 | .2500 | .5000 | .7500 | .9000 | .9500 | .9750 | .9900 | .9950 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3.93e-5 | .000157 | .000982 | .00393 | .0158 | .102 | .455 | 1.32 | 2.71 | 3.84 | 5.02 | 6.63 | 7.88 |
| 2 | .0100 | .0201 | .0506 | .103 | .211 | .575 | 1.39 | 2.77 | 4.61 | 5.99 | 7.38 | 9.21 | 10.6 |
| 3 | .0717 | .115 | .216 | .352 | .584 | 1.21 | 2.37 | 4.11 | 6.25 | 7.81 | 9.35 | 11.3 | 12.8 |
| 4 | .207 | .297 | .484 | .711 | 1.06 | 1.92 | 3.36 | 5.39 | 7.78 | 9.49 | 11.1 | 13.3 | 14.9 |
| 5 | .412 | .554 | .831 | 1.15 | 1.61 | 2.67 | 4.35 | 6.63 | 9.24 | 11.1 | 12.8 | 15.1 | 16.7 |
| 6 | .676 | .872 | 1.24 | 1.64 | 2.20 | 3.45 | 5.35 | 7.84 | 10.6 | 12.6 | 14.4 | 16.8 | 18.5 |
| 7 | .989 | 1.24 | 1.69 | 2.17 | 2.83 | 4.25 | 6.35 | 9.04 | 12.0 | 14.1 | 16.0 | 18.5 | 20.3 |
| 8 | 1.34 | 1.65 | 2.18 | 2.73 | 3.49 | 5.07 | 7.34 | 10.2 | 13.4 | 15.5 | 17.5 | 20.1 | 22.0 |
| 9 | 1.73 | 2.09 | 2.70 | 3.33 | 4.17 | 5.90 | 8.34 | 11.4 | 14.7 | 16.9 | 19.0 | 21.7 | 23.6 |
| 10 | 2.16 | 2.56 | 3.25 | 3.94 | 4.87 | 6.74 | 9.34 | 12.5 | 16.0 | 18.3 | 20.5 | 23.2 | 25.2 |
| 11 | 2.60 | 3.05 | 3.82 | 4.57 | 5.58 | 7.58 | 10.3 | 13.7 | 17.3 | 19.7 | 21.9 | 24.7 | 26.8 |
| 12 | 3.07 | 3.57 | 4.40 | 5.23 | 6.30 | 8.44 | 11.3 | 14.8 | 18.5 | 21.0 | 23.3 | 26.2 | 28.3 |
| 13 | 3.57 | 4.11 | 5.01 | 5.89 | 7.04 | 9.30 | 12.3 | 16.0 | 19.8 | 22.4 | 24.7 | 27.7 | 29.8 |
| 14 | 4.07 | 4.66 | 5.63 | 6.57 | 7.79 | 10.2 | 13.3 | 17.1 | 21.1 | 23.7 | 26.1 | 29.1 | 31.3 |
| 15 | 4.60 | 5.23 | 6.26 | 7.26 | 8.55 | 11.0 | 14.3 | 18.2 | 22.3 | 25.0 | 27.5 | 30.6 | 32.8 |
| 16 | 5.14 | 5.81 | 6.91 | 7.96 | 9.31 | 11.9 | 15.3 | 19.4 | 23.5 | 26.3 | 28.8 | 32.0 | 34.3 |
| 17 | 5.70 | 6.41 | 7.56 | 8.67 | 10.1 | 12.8 | 16.3 | 20.5 | 24.8 | 27.6 | 30.2 | 33.4 | 35.7 |
| 18 | 6.26 | 7.01 | 8.23 | 9.39 | 10.9 | 13.7 | 17.3 | 21.6 | 26.0 | 28.9 | 31.5 | 34.8 | 37.2 |
| 19 | 6.84 | 7.63 | 8.91 | 10.1 | 11.7 | 14.6 | 18.3 | 22.7 | 27.2 | 30.1 | 32.9 | 36.2 | 38.6 |
| 20 | 7.43 | 8.26 | 9.59 | 10.9 | 12.4 | 15.5 | 19.3 | 23.8 | 28.4 | 31.4 | 34.2 | 37.6 | 40.0 |
| 21 | 8.03 | 8.90 | 10.3 | 11.6 | 13.2 | 16.3 | 20.3 | 24.9 | 29.6 | 32.7 | 35.5 | 38.9 | 41.4 |
| 22 | 8.64 | 9.54 | 11.0 | 12.3 | 14.0 | 17.2 | 21.3 | 26.0 | 30.8 | 33.9 | 36.8 | 40.3 | 42.8 |
| 23 | 9.26 | 10.2 | 11.7 | 13.1 | 14.8 | 18.1 | 22.3 | 27.1 | 32.0 | 35.2 | 38.1 | 41.6 | 44.2 |
| 24 | 9.89 | 10.9 | 12.4 | 13.8 | 15.7 | 19.0 | 23.3 | 28.2 | 33.2 | 36.4 | 39.4 | 43.0 | 45.6 |
| 25 | 10.5 | 11.5 | 13.1 | 14.6 | 16.5 | 19.9 | 24.3 | 29.3 | 34.4 | 37.7 | 40.6 | 44.3 | 46.9 |
| 26 | 11.2 | 12.2 | 13.8 | 15.4 | 17.3 | 20.8 | 25.3 | 30.4 | 35.6 | 38.9 | 41.9 | 45.6 | 48.3 |
| 27 | 11.8 | 12.9 | 14.6 | 16.2 | 18.1 | 21.7 | 26.3 | 31.5 | 36.7 | 40.1 | 43.2 | 47.0 | 49.6 |
| 28 | 12.5 | 13.6 | 15.3 | 16.9 | 18.9 | 22.7 | 27.3 | 32.6 | 37.9 | 41.3 | 44.5 | 48.3 | 51.0 |
| 29 | 13.1 | 14.3 | 16.0 | 17.7 | 19.8 | 23.6 | 28.3 | 33.7 | 39.1 | 42.6 | 45.7 | 49.6 | 52.3 |
| 30 | 13.8 | 15.0 | 16.8 | 18.5 | 20.6 | 24.5 | 29.3 | 34.8 | 40.3 | 43.8 | 47.0 | 50.9 | 53.7 |

5. Случајна променљива $Z$ има нормалну $\mathcal{N}(0,1)$ расподелу и нека је $X = Z^2$. Израчунати вероватноћу $P(X < 2{,}25)$ и наћи вредност $x$ за коју је $P(X > x) = 0{,}95$.

$Z_1, Z_2, \ldots, Z_n$ независне случајне променљиве, $Z_i : \mathcal{N}(0,1)$, $i = 1,2,\ldots,n$
ако је $X = \sum_{i=1}^{n} Z_i^2$, онда $X$ има Пирсонову $\chi^2$ расподелу, $X : \chi^2_n$ n-број степени слободе

$\Rightarrow X : \chi^2_1$

$P(X < 2{,}25) \approx 0{,}9$ R: 0,8664

$P(X > x) = 0{,}95 \Rightarrow P(X \leq x) = 0{,}05$

## 4.3 Pirsonova $\chi^2$ raspodela $F(y) = \int_0^y (x^{\frac{n}{2}-1} e^{-\frac{x}{2}})/(2^{\frac{n}{2}}\Gamma(n/2))dx$

| $n$ \ $F$ | .0050 | .0100 | .0250 | .0500 | .1000 | .2500 | .5000 | .7500 | .9000 | .9500 | .9750 | .9900 | .9950 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3.93e-5 | .000157 | .000982 | .00393 | .0158 | .102 | .455 | 1.32 | 2.71 | 3.84 | 5.02 | 6.63 | 7.88 |
| 2 | .0100 | .0201 | .0506 | .103 | .211 | .575 | 1.39 | 2.77 | 4.61 | 5.99 | 7.38 | 9.21 | 10.6 |
| 3 | .0717 | .115 | .216 | .352 | .584 | 1.21 | 2.37 | 4.11 | 6.25 | 7.81 | 9.35 | 11.3 | 12.8 |
| 4 | .207 | .297 | .484 | .711 | 1.06 | 1.92 | 3.36 | 5.39 | 7.78 | 9.49 | 11.1 | 13.3 | 14.9 |
| 5 | .412 | .554 | .831 | 1.15 | 1.61 | 2.67 | 4.35 | 6.63 | 9.24 | 11.1 | 12.8 | 15.1 | 16.7 |
| 6 | .676 | .872 | 1.24 | 1.64 | 2.20 | 3.45 | 5.35 | 7.84 | 10.6 | 12.6 | 14.4 | 16.8 | 18.5 |
| 7 | .989 | 1.24 | 1.69 | 2.17 | 2.83 | 4.25 | 6.35 | 9.04 | 12.0 | 14.1 | 16.0 | 18.5 | 20.3 |
| 8 | 1.34 | 1.65 | 2.18 | 2.73 | 3.49 | 5.07 | 7.34 | 10.2 | 13.4 | 15.5 | 17.5 | 20.1 | 22.0 |
| 9 | 1.73 | 2.09 | 2.70 | 3.33 | 4.17 | 5.90 | 8.34 | 11.4 | 14.7 | 16.9 | 19.0 | 21.7 | 23.6 |
| 10 | 2.16 | 2.56 | 3.25 | 3.94 | 4.87 | 6.74 | 9.34 | 12.5 | 16.0 | 18.3 | 20.5 | 23.2 | 25.2 |
| 11 | 2.60 | 3.05 | 3.82 | 4.57 | 5.58 | 7.58 | 10.3 | 13.7 | 17.3 | 19.7 | 21.9 | 24.7 | 26.8 |
| 12 | 3.07 | 3.57 | 4.40 | 5.23 | 6.30 | 8.44 | 11.3 | 14.8 | 18.5 | 21.0 | 23.3 | 26.2 | 28.3 |
| 13 | 3.57 | 4.11 | 5.01 | 5.89 | 7.04 | 9.30 | 12.3 | 16.0 | 19.8 | 22.4 | 24.7 | 27.7 | 29.8 |
| 14 | 4.07 | 4.66 | 5.63 | 6.57 | 7.79 | 10.2 | 13.3 | 17.1 | 21.1 | 23.7 | 26.1 | 29.1 | 31.3 |
| 15 | 4.60 | 5.23 | 6.26 | 7.26 | 8.55 | 11.0 | 14.3 | 18.2 | 22.3 | 25.0 | 27.5 | 30.6 | 32.8 |
| 16 | 5.14 | 5.81 | 6.91 | 7.96 | 9.31 | 11.9 | 15.3 | 19.4 | 23.5 | 26.3 | 28.8 | 32.0 | 34.3 |
| 17 | 5.70 | 6.41 | 7.56 | 8.67 | 10.1 | 12.8 | 16.3 | 20.5 | 24.8 | 27.6 | 30.2 | 33.4 | 35.7 |
| 18 | 6.26 | 7.01 | 8.23 | 9.39 | 10.9 | 13.7 | 17.3 | 21.6 | 26.0 | 28.9 | 31.5 | 34.8 | 37.2 |
| 19 | 6.84 | 7.63 | 8.91 | 10.1 | 11.7 | 14.6 | 18.3 | 22.7 | 27.2 | 30.1 | 32.9 | 36.2 | 38.6 |
| 20 | 7.43 | 8.26 | 9.59 | 10.9 | 12.4 | 15.5 | 19.3 | 23.8 | 28.4 | 31.4 | 34.2 | 37.6 | 40.0 |
| 21 | 8.03 | 8.90 | 10.3 | 11.6 | 13.2 | 16.3 | 20.3 | 24.9 | 29.6 | 32.7 | 35.5 | 38.9 | 41.4 |
| 22 | 8.64 | 9.54 | 11.0 | 12.3 | 14.0 | 17.2 | 21.3 | 26.0 | 30.8 | 33.9 | 36.8 | 40.3 | 42.8 |
| 23 | 9.26 | 10.2 | 11.7 | 13.1 | 14.8 | 18.1 | 22.3 | 27.1 | 32.0 | 35.2 | 38.1 | 41.6 | 44.2 |
| 24 | 9.89 | 10.9 | 12.4 | 13.8 | 15.7 | 19.0 | 23.3 | 28.2 | 33.2 | 36.4 | 39.4 | 43.0 | 45.6 |
| 25 | 10.5 | 11.5 | 13.1 | 14.6 | 16.5 | 19.9 | 24.3 | 29.3 | 34.4 | 37.7 | 40.6 | 44.3 | 46.9 |
| 26 | 11.2 | 12.2 | 13.8 | 15.4 | 17.3 | 20.8 | 25.3 | 30.4 | 35.6 | 38.9 | 41.9 | 45.6 | 48.3 |
| 27 | 11.8 | 12.9 | 14.6 | 16.2 | 18.1 | 21.7 | 26.3 | 31.5 | 36.7 | 40.1 | 43.2 | 47.0 | 49.6 |
| 28 | 12.5 | 13.6 | 15.3 | 16.9 | 18.9 | 22.7 | 27.3 | 32.6 | 37.9 | 41.3 | 44.5 | 48.3 | 51.0 |
| 29 | 13.1 | 14.3 | 16.0 | 17.7 | 19.8 | 23.6 | 28.3 | 33.7 | 39.1 | 42.6 | 45.7 | 49.6 | 52.3 |
| 30 | 13.8 | 15.0 | 16.8 | 18.5 | 20.6 | 24.5 | 29.3 | 34.8 | 40.3 | 43.8 | 47.0 | 50.9 | 53.7 |

5. Случајна променљива $Z$ има нормалну $N(0,1)$ расподелу и нека је $X = Z^2$. Израчунати вероватноћу $P(X < 2{,}25)$ и наћи вредност $x$ за коју је $P(X > x) = 0{,}95$.

$Z_1, Z_2, \ldots, Z_n$ независне случајне променљиве, $Z_i : N(0,1)$, $i = 1, 2, \ldots, n$

ако је $X = \sum_{i=1}^{n} Z_i^2$, онда $X$ има Пирсонову $\chi^2$ расподелу, $X : \chi^2_n$ — $n$-број степени слободе

$\Rightarrow X : \chi^2_1$

$P(X < 2{,}25) \approx 0{,}9$ R: 0,8664

$P(X > x) = 0{,}95 \Rightarrow P(X \leq x) = 0{,}05$

$x = 0{,}00393$

## 4.3 Pirsonova $\chi^2$ raspodela $F(y) = \int_0^y (x^{\frac{n}{2}-1} e^{-\frac{x}{2}})/(2^{\frac{n}{2}}\Gamma(n/2))dx$

| $n$ \ $F$ | .0050 | .0100 | .0250 | .0500 | .1000 | .2500 | .5000 | .7500 | .9000 | .9500 | .9750 | .9900 | .9950 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3.93e-5 | .000157 | .000982 | .00393 | .0158 | .102 | .455 | 1.32 | 2.71 | 3.84 | 5.02 | 6.63 | 7.88 |
| 2 | .0100 | .0201 | .0506 | .103 | .211 | .575 | 1.39 | 2.77 | 4.61 | 5.99 | 7.38 | 9.21 | 10.6 |
| 3 | .0717 | .115 | .216 | .352 | .584 | 1.21 | 2.37 | 4.11 | 6.25 | 7.81 | 9.35 | 11.3 | 12.8 |
| 4 | .207 | .297 | .484 | .711 | 1.06 | 1.92 | 3.36 | 5.39 | 7.78 | 9.49 | 11.1 | 13.3 | 14.9 |
| 5 | .412 | .554 | .831 | 1.15 | 1.61 | 2.67 | 4.35 | 6.63 | 9.24 | 11.1 | 12.8 | 15.1 | 16.7 |
| 6 | .676 | .872 | 1.24 | 1.64 | 2.20 | 3.45 | 5.35 | 7.84 | 10.6 | 12.6 | 14.4 | 16.8 | 18.5 |
| 7 | .989 | 1.24 | 1.69 | 2.17 | 2.83 | 4.25 | 6.35 | 9.04 | 12.0 | 14.1 | 16.0 | 18.5 | 20.3 |
| 8 | 1.34 | 1.65 | 2.18 | 2.73 | 3.49 | 5.07 | 7.34 | 10.2 | 13.4 | 15.5 | 17.5 | 20.1 | 22.0 |
| 9 | 1.73 | 2.09 | 2.70 | 3.33 | 4.17 | 5.90 | 8.34 | 11.4 | 14.7 | 16.9 | 19.0 | 21.7 | 23.6 |
| 10 | 2.16 | 2.56 | 3.25 | 3.94 | 4.87 | 6.74 | 9.34 | 12.5 | 16.0 | 18.3 | 20.5 | 23.2 | 25.2 |
| 11 | 2.60 | 3.05 | 3.82 | 4.57 | 5.58 | 7.58 | 10.3 | 13.7 | 17.3 | 19.7 | 21.9 | 24.7 | 26.8 |
| 12 | 3.07 | 3.57 | 4.40 | 5.23 | 6.30 | 8.44 | 11.3 | 14.8 | 18.5 | 21.0 | 23.3 | 26.2 | 28.3 |
| 13 | 3.57 | 4.11 | 5.01 | 5.89 | 7.04 | 9.30 | 12.3 | 16.0 | 19.8 | 22.4 | 24.7 | 27.7 | 29.8 |
| 14 | 4.07 | 4.66 | 5.63 | 6.57 | 7.79 | 10.2 | 13.3 | 17.1 | 21.1 | 23.7 | 26.1 | 29.1 | 31.3 |
| 15 | 4.60 | 5.23 | 6.26 | 7.26 | 8.55 | 11.0 | 14.3 | 18.2 | 22.3 | 25.0 | 27.5 | 30.6 | 32.8 |
| 16 | 5.14 | 5.81 | 6.91 | 7.96 | 9.31 | 11.9 | 15.3 | 19.4 | 23.5 | 26.3 | 28.8 | 32.0 | 34.3 |
| 17 | 5.70 | 6.41 | 7.56 | 8.67 | 10.1 | 12.8 | 16.3 | 20.5 | 24.8 | 27.6 | 30.2 | 33.4 | 35.7 |
| 18 | 6.26 | 7.01 | 8.23 | 9.39 | 10.9 | 13.7 | 17.3 | 21.6 | 26.0 | 28.9 | 31.5 | 34.8 | 37.2 |
| 19 | 6.84 | 7.63 | 8.91 | 10.1 | 11.7 | 14.6 | 18.3 | 22.7 | 27.2 | 30.1 | 32.9 | 36.2 | 38.6 |
| 20 | 7.43 | 8.26 | 9.59 | 10.9 | 12.4 | 15.5 | 19.3 | 23.8 | 28.4 | 31.4 | 34.2 | 37.6 | 40.0 |
| 21 | 8.03 | 8.90 | 10.3 | 11.6 | 13.2 | 16.3 | 20.3 | 24.9 | 29.6 | 32.7 | 35.5 | 38.9 | 41.4 |
| 22 | 8.64 | 9.54 | 11.0 | 12.3 | 14.0 | 17.2 | 21.3 | 26.0 | 30.8 | 33.9 | 36.8 | 40.3 | 42.8 |
| 23 | 9.26 | 10.2 | 11.7 | 13.1 | 14.8 | 18.1 | 22.3 | 27.1 | 32.0 | 35.2 | 38.1 | 41.6 | 44.2 |
| 24 | 9.89 | 10.9 | 12.4 | 13.8 | 15.7 | 19.0 | 23.3 | 28.2 | 33.2 | 36.4 | 39.4 | 43.0 | 45.6 |
| 25 | 10.5 | 11.5 | 13.1 | 14.6 | 16.5 | 19.9 | 24.3 | 29.3 | 34.4 | 37.7 | 40.6 | 44.3 | 46.9 |
| 26 | 11.2 | 12.2 | 13.8 | 15.4 | 17.3 | 20.8 | 25.3 | 30.4 | 35.6 | 38.9 | 41.9 | 45.6 | 48.3 |
| 27 | 11.8 | 12.9 | 14.6 | 16.2 | 18.1 | 21.7 | 26.3 | 31.5 | 36.7 | 40.1 | 43.2 | 47.0 | 49.6 |
| 28 | 12.5 | 13.6 | 15.3 | 16.9 | 18.9 | 22.7 | 27.3 | 32.6 | 37.9 | 41.3 | 44.5 | 48.3 | 51.0 |
| 29 | 13.1 | 14.3 | 16.0 | 17.7 | 19.8 | 23.6 | 28.3 | 33.7 | 39.1 | 42.6 | 45.7 | 49.6 | 52.3 |
| 30 | 13.8 | 15.0 | 16.8 | 18.5 | 20.6 | 24.5 | 29.3 | 34.8 | 40.3 | 43.8 | 47.0 | 50.9 | 53.7 |

# ДВОДИМЕНЗИОНАЛНЕ СЛУЧАЈНЕ ПРОМЕНЉИВЕ

$(X,Y)$ је **дводимензионална случајна променљива** над простором вероватноће $(\Omega,\mathcal{F},P)$ ако су $X$ и $Y$ случајне променљиве над $(\Omega,\mathcal{F},P)$

$$F(x,y)=P(X\leq x, Y\leq y)$$

6. Дводимензионална случајна променљива $(X,Y)$ дата је законом расподеле

| $X \setminus Y$ | 0 | 1 |
|---|---|---|
| 0 | $\frac{1}{18}$ | $\frac{6}{18}$ |
| 1 | $\frac{3}{18}$ | $\frac{4}{18}$ |
| 2 | $\frac{3}{18}$ | $\frac{1}{18}$ |

а) Наћи маргиналне расподеле за случајне променљиве $X$ и $Y$

$P(X=0) = P(X=0,Y=0) + P(X=0,Y=1) = \frac{1}{18} + \frac{6}{18} = \frac{7}{18}$

$P(X=1) = \frac{3}{18} + \frac{4}{18} = \frac{7}{18}$

$P(X=2) = \frac{3}{18} + \frac{1}{18} = \frac{4}{18}$

$$X: \begin{pmatrix} 0 & 1 & 2 \\ \frac{7}{18} & \frac{7}{18} & \frac{4}{18} \end{pmatrix}$$

| $X \setminus Y$ | 0 | 1 | |
|---|---|---|---|
| 0 | $\frac{1}{18}$ | $\frac{6}{18}$ | $\frac{7}{18}$ |
| 1 | $\frac{3}{18}$ | $\frac{4}{18}$ | $\frac{7}{18}$ |
| 2 | $\frac{3}{18}$ | $\frac{1}{18}$ | $\frac{4}{18}$ |
| | $\frac{7}{18}$ | $\frac{11}{18}$ | 1 |

$P(Y=0) = P(X=0,Y=0) + P(X=1,Y=0) + P(X=2,Y=0) = \frac{1}{18} + \frac{3}{18} + \frac{3}{18} = \frac{7}{18}$

$P(Y=1) = \frac{6}{18} + \frac{4}{18} + \frac{1}{18} = \frac{11}{18}$

$$Y: \begin{pmatrix} 0 & 1 \\ \frac{7}{18} & \frac{11}{18} \end{pmatrix}$$

б) Испитати независност случајних променљивих $X$ и $Y$

Дискретне случајне променљиве $X$ и $Y$ су независне ако је $p_{ij} = p_{i\cdot} \cdot p_{\cdot j}$

$P(X=0,Y=0) = \frac{1}{18} \neq \frac{7}{18} \cdot \frac{7}{18} = P(X=0) \cdot P(Y=0)$

$\Rightarrow$ $X$ и $Y$ нису независне, тј. зависне су

Напомена:

$p_{ij} = p_{i\cdot} \cdot p_{\cdot j}$ је скраћени запис за

$P(X=i, Y=j) = P(X=i) \cdot P(Y=j)$

| X\|Y | 0 | 1 | |
|---|---|---|---|
| 0 | 1/18 | 6/18 | 7/18 |
| 1 | 3/18 | 4/18 | 7/18 |
| 2 | 5/18 | 1/18 | 4/18 |
| | 7/18 | 11/18 | |

б) Израчунати веројатноќе

$P(X>0) = P(X=1) + P(X=2) = \frac{7}{18} + \frac{4}{18} = \frac{11}{18}$

$P(Y \le 2) = 1$

$P(X<1, Y \le 2) = P(X=0, Y=0) + P(X=0, Y=1) = \frac{7}{18}$

УСЛОВНЕ ВЕРОЈАТНОЋЕ

$p(x_i|y_j) = \frac{p_{ij}}{p_{\cdot j}}$ $\quad p(y_j|x_i) = \frac{p_{ij}}{p_{i\cdot}}$

$P(X=1|Y=1) = \frac{P(X=1, Y=1)}{P(Y=1)} = \frac{\frac{4}{18}}{\frac{11}{18}} = \frac{4}{11}$

$P(Y=1|X=2) = \frac{P(X=2, Y=1)}{P(X=2)} = \frac{\frac{1}{18}}{\frac{4}{18}} = \frac{1}{4}$

г) Израчунати

$F_{X,Y}(-1,0) = P(X \le -1, Y \le 0) = 0$

$F_{X,Y}(1, 1.35) = P(X \le 1, Y \le 1.35) = P(X=0, Y=0) + P(X=0, Y=1) + P(X=1, Y=0) + P(X=1, Y=1)$

$= \frac{1}{18} + \frac{6}{18} + \frac{3}{18} + \frac{4}{18} = \frac{14}{18}$

$F_{X,Y}(2,2) = P(X \le 2, Y \le 2) = 1$

7. Бацa се коцкица за игру 2 пута. Нека је $X$ број појављивања броја дељивог са 3, а $Y$ број појављивања броја 3. Наћи закон расподеле случајне променљиве $(X,Y)$ и испитати независност случајних променљивих $X$ и $Y$.

$P(ij)=\frac{1}{36}$

| $X\backslash Y$ | 0 | 1 | 2 | |
|---|---|---|---|---|
| 0 | $\frac{16}{36}$ | 0 | 0 | $\frac{16}{36}$ |
| 1 | $\frac{8}{36}$ | $\frac{8}{36}$ | 0 | $\frac{16}{36}$ |
| 2 | $\frac{1}{36}$ | $\frac{2}{36}$ | $\frac{1}{36}$ | $\frac{4}{36}$ |
| | $\frac{25}{36}$ | $\frac{10}{36}$ | $\frac{1}{36}$ | |

$P(X=0,Y=0)=P(11+12+14+15+21+22+24+25+41+42+44+45+51+52+54+55)=\frac{16}{36}$

$P(X=0,Y=1)=0$

$P(X=0,Y=2)=0$

$P(X=1,Y=0)=P(16+26+46+56+61+62+64+65)=\frac{8}{36}$

$P(X=1,Y=1)=P(13+23+31+32+34+35+43+53)=\frac{8}{36}$

$P(X=1,Y=2)=0$

$P(X=2,Y=0)=P(66)=\frac{1}{36}$

$P(X=2,Y=1)=P(36+63)=\frac{2}{36}$

$P(X=2,Y=2)=P(33)=\frac{1}{36}$

$P(X=0,Y=1)=0$

$P(X=0)=\frac{16}{36}$ $P(Y=1)=\frac{10}{36}$

$\Rightarrow$ $X$ и $Y$ нису независне случајне променљиве

8. У једној кутији се налазе куглице означене бројевима 1,2,3,4, а у другој кутији куглице означене бројевима 2,4,6, и из сваке кутије се извлачи по једна куглица. Нека случајна променљива X представља број на извученој куглици из прве кутије, а Y број на извученој куглици из друге кутије

а) Наћи закон расподеле дводимензионалне случајне променљиве $(X,Y)$

| X \ Y | 2 | 4 | 6 | |
|---|---|---|---|---|
| 1 | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{4}$ |
| 2 | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{4}$ |
| 3 | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{4}$ |
| 4 | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{4}$ |
| | $\frac{1}{3}$ | $\frac{1}{3}$ | $\frac{1}{3}$ | |

$P(ij) = \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{12}$

$i = 1,2,3,4 \quad j = 2,4,6$

б) Наћи маргиналне случајне променљиве X и Y и утврдити да ли су независне

$X: \begin{pmatrix} 1 & 2 & 3 & 4 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix} \qquad Y: \begin{pmatrix} 2 & 4 & 6 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}$

$P(X=i) = \frac{1}{4}$

$P(Y=j) = \frac{1}{3}$

$P(X=i, Y=j) = \frac{1}{12} \ \forall i,j \implies$ X и Y су независне случајне променљиве

б) Наћи закон расподеле дводимензионалне случајне променљиве $(U,V)$, где је $U=2X$ и $V=\min\{X,Y\}$.

$R_U=\{2,4,6,8\}$ $R_V=\{1,2,3,4\}$

| U\V | 1 | 2 | 3 | 4 | |
|---|---|---|---|---|---|
| 2 | $\frac{3}{12}$ | 0 | 0 | 0 | $\frac{3}{12}$ |
| 4 | 0 | $\frac{3}{12}$ | 0 | 0 | $\frac{3}{12}$ |
| 6 | 0 | $\frac{1}{12}$ | $\frac{2}{12}$ | 0 | $\frac{3}{12}$ |
| 8 | 0 | $\frac{1}{12}$ | 0 | $\frac{2}{12}$ | $\frac{3}{12}$ |
| | $\frac{3}{12}$ | $\frac{5}{12}$ | $\frac{2}{12}$ | $\frac{2}{12}$ | |

$P(U=2,V=1)=P(X=1)=\frac{3}{12}$

$P(U=2,V=2)=P(U=2,V=3)=P(U=2,V=4)=0$

$P(U=4,V=1)=0$

$P(U=4,V=2)=P(X=2)=\frac{3}{12}$

$P(U=4,V=3)=P(U=4,V=4)=0$

$P(U=6,V=1)=0$

$P(U=6,V=2)=P(X=3,Y=2)=\frac{1}{12}$

$P(U=6,V=3)=P(X=3,Y=4)+P(X=3,Y=6)=\frac{2}{12}$

$P(U=6,V=4)=0$

$P(U=8,V=1)=0$

$P(U=8,V=2)=P(X=4,Y=2)=\frac{1}{12}$

$P(U=8,V=3)=0$

$P(U=8,V=4)=P(X=4,Y=4)+P(X=4,Y=6)=\frac{2}{12}$

г) Утврдити да ли су маргиналне случајне променљиве $U$ и $V$ независне.

$$\left.\begin{array}{l} P(U=2, V=2) = 0 \\ P(U=2) = \frac{3}{12},\ P(V=2) = \frac{5}{12} \end{array}\right\} \Rightarrow U \text{ и } V \text{ нису независне}$$

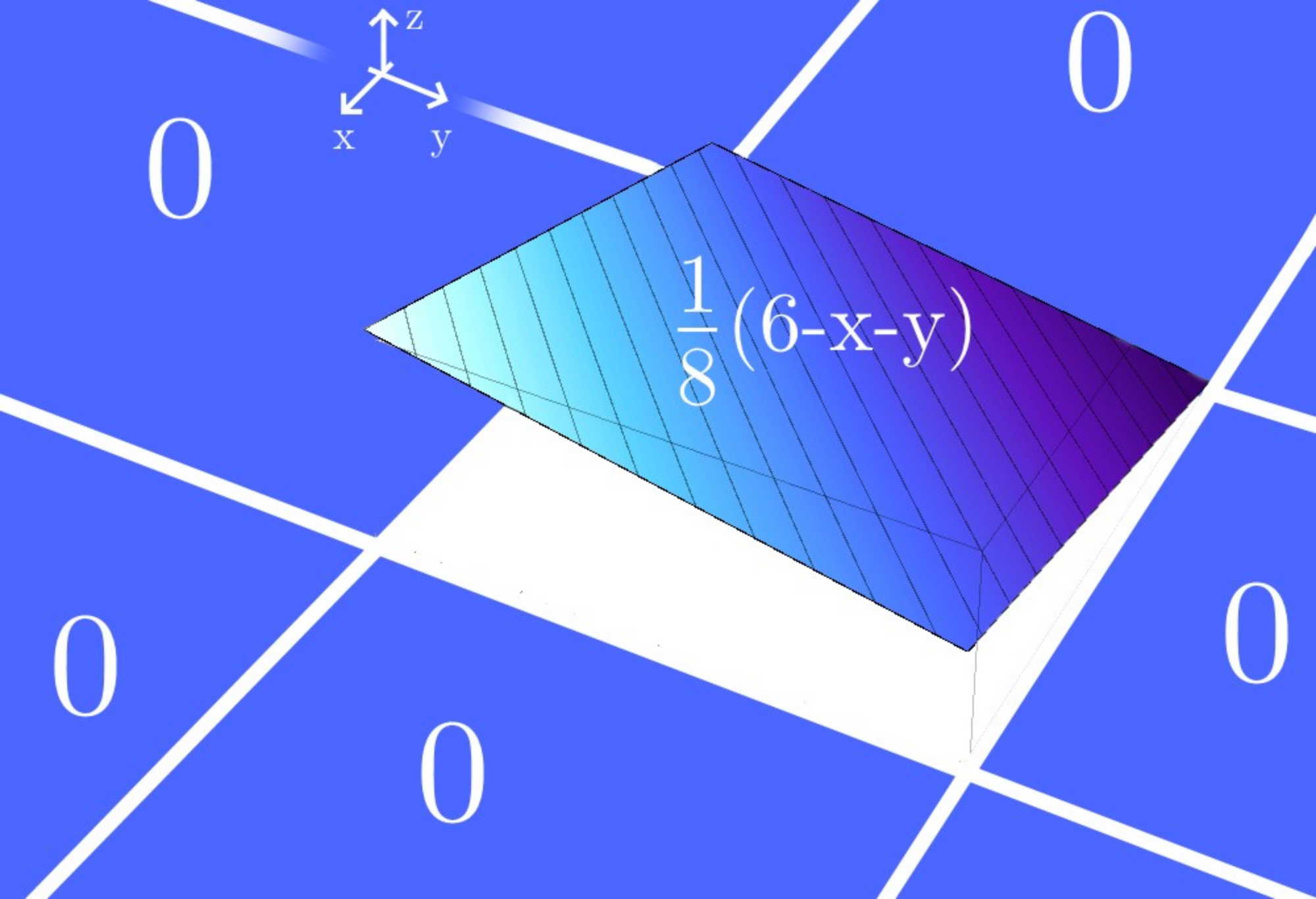
д) Израчунати вероватноће

$$P(X \leq 3, Y \leq 2) = P(X=1, Y=2) + P(X=2, Y=2) + P(X=3, Y=2) = \frac{3}{12}$$

$$P(U \leq 4, V \leq 2) = P(U=2, V=1) + P(U=2, V=2) + P(U=4, V=1) + P(U=4, V=2) = \frac{6}{12} = \frac{1}{2}$$

9. Непрекидна случајна променљива $(X,Y)$ дата је густином $\varphi_{X,Y}(x,y)=\begin{cases}\frac{1}{8}(6-x-y), & x\in[0,2],\ y\in[2,4]\\ 0, & \text{иначе}\end{cases}$

а) Израчунати вероватноће $P(X<1, 2.5<Y<3.5)$ и $P(X>1, Y>3)$


z
x
y
0
0
0
0
0
$\frac{1}{8}(6\text{-}x\text{-}y)$

9. Непрекидна случајна променљива $(X,Y)$ дата је густином $\varphi_{X,Y}(x,y)=\begin{cases}\frac{1}{8}(6-x-y), & x\in[0,2],\ y\in[2,4]\\ 0, & \text{иначе}\end{cases}$

а) Израчунати вероватноће $P(X<1, 2,5<Y<3,5)$ и $P(X>1, Y>3)$

$$P(X<1, 2,5<Y<3,5)=\int_0^1\int_{2,5}^{3,5}\frac{1}{8}(6-x-y)\,dy\,dx=\frac{1}{8}\int_0^1\left(6y-xy-\frac{y^2}{2}\right)\Big|_{2,5}^{3,5}dx=\frac{1}{8}\int_0^1(3-x)\,dx$$

Прво интегралимо по $y$, а $x$ посматрамо као константу

$$=\frac{1}{8}\left(3x-\frac{x^2}{2}\right)\Big|_0^1=0,3125$$

$$P(X>1,Y>3)=\int_1^2\int_3^4\frac{1}{8}(6-x-y)\,dy\,dx=\int_3^4\int_1^2\frac{1}{8}(6-x-y)\,dx\,dy=\frac{1}{8}\int_3^4\left(6x-\frac{x^2}{2}-xy\right)\Big|_1^2dy$$

Сада ћемо прво интегралити по $x$, $y$ третирамо као константу

$$=\frac{1}{8}\int_3^4\left(\frac{9}{2}-y\right)dy=\frac{1}{8}\left(\frac{9}{2}y-\frac{y^2}{2}\right)\Big|_3^4=0,125$$

б) Израчунати $F_{X,Y}(1{,}2, 3{,}2)$, $F_{X,Y}(-1,-2)$ и $F_{X,Y}(6,6)$

$$F_{X,Y}(1{,}2, 3{,}2) = \int_{-\infty}^{1{,}2}\int_{-\infty}^{3{,}2} \varphi_{X,Y}(x,y)\,dy\,dx = \int_{0}^{1{,}2}\int_{2}^{3{,}2} \frac{1}{8}(6-x-y)\,dy\,dx = \ldots = 0{,}312$$

$$F_{X,Y}(-1,-2) = \int_{-\infty}^{-1}\int_{-\infty}^{-2} \varphi_{X,Y}(x,y)\,dy\,dx = \int_{-\infty}^{-1}\int_{-\infty}^{-2} 0\,dy\,dx = 0$$

$$F_{X,Y}(6,6) = \int_{-\infty}^{6}\int_{-\infty}^{6} \varphi_{X,Y}(x,y)\,dy\,dx = \int_{0}^{2}\int_{2}^{4} \frac{1}{8}(6-x-y)\,dy\,dx = 1^{*}$$

* За задатак проверити да је вредност овог двоструког интеграла 1

б) Наћи маргиналне расподеле за $X$ и $Y$ и испитати њихову независност.

$x \in [0,2]$

$$\varphi_X(x) = \int_{-\infty}^{\infty} \varphi_{X,Y}(x,y)\,dy = \int_2^4 \frac{1}{8}(6-x-y)\,dy = \frac{1}{8}\left(6y - xy - \frac{y^2}{2}\right)\Big|_2^4 = \frac{1}{8}(6-2x) = \frac{1}{4}(3-x)$$

$y \in [2,4]$

$$\varphi_Y(y) = \int_{-\infty}^{\infty} \varphi_{X,Y}(x,y)\,dx = \int_0^2 \frac{1}{8}(6-x-y)\,dx = \frac{1}{8}\left(6x - \frac{x^2}{2} - xy\right)\Big|_0^2 = \frac{1}{8}(10-2y) = \frac{1}{4}(5-y)$$

Непрекидне случајне променљиве $X$ и $Y$ су независне ако је $\varphi(x,y) = \varphi_X(x) \cdot \varphi_Y(y)$

$$\varphi_X(x) \cdot \varphi_Y(y) = \frac{1}{16}(3-x)(5-y) = \frac{15-3y-5x+xy}{16} \neq \frac{6-x-y}{8} = \varphi_{X,Y}(x,y)$$

$\Rightarrow$ $X$ и $Y$ нису независне

* Независне случајне променљиве $X$ и $Y$ дате су густинама

$$\varphi_X(x)=\begin{cases}\frac{1}{x}, & x\in[1,e]\\ 0, & x\notin[1,e]\end{cases} \qquad \varphi_Y(y)=\begin{cases}2(1-y), & y\in[0,1]\\ 0, & y\notin[0,1]\end{cases}$$

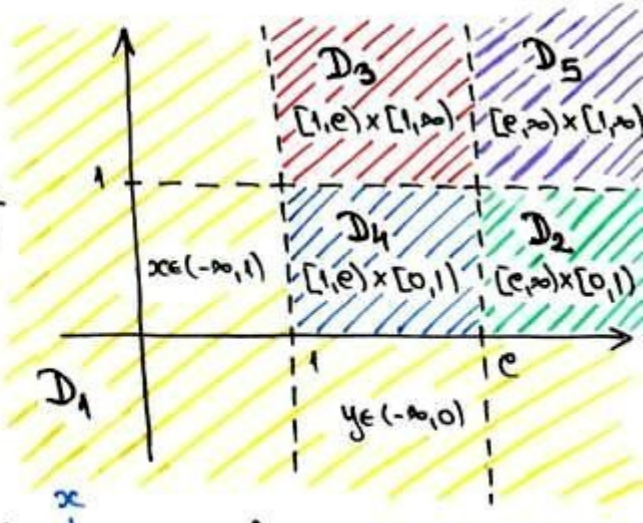
За студенте који желе да знају више ☺

Наћи функцију расподеле дводимензионалне случајне променљиве $(X,Y)$.

$X$ и $Y$ независне $\Rightarrow \varphi_X\cdot\varphi_Y=\varphi_{X,Y}$

$$\varphi_{X,Y}(x,y)=\begin{cases}\frac{2(1-y)}{x}, & (x,y)\in[1,e]\times[0,1]\\ 0, & (x,y)\notin[1,e]\times[0,1]\end{cases}$$

$x\to u$
$y\to v$




- $(x,y)\in D_1 \quad (x<1 \vee y<0)$

$$F_{X,Y}(x,y)=\int_{-\infty}^{x}\int_{-\infty}^{y}0\,dy\,dx=0$$

- $(x,y)\in D_2 \quad (x\geq e \wedge y\in[0,1))$

$$F_{X,Y}(x,y)=\int_1^e\int_0^y\frac{2(1-v)}{x}\,dv\,dx=2\int_0^y(1-v)\ln x\Big|_1^e dv=2\int_0^y(1-v)\,dv=2\left(v-\frac{v^2}{2}\right)\Big|_0^y=2y-y^2$$

- $(x,y)\in D_3 \quad (x\in[1,e) \wedge y\geq 1)$

$$F_{X,Y}(x,y)=\int_1^x\int_0^1\frac{2(1-y)}{u}\,dy\,du=\int_1^x\frac{2}{u}\left(y-\frac{y^2}{2}\right)\Big|_0^1 du=\int_1^x\frac{2}{u}\cdot\frac{1}{2}\,du=\ln u\Big|_1^x=\ln x$$

- $(x,y)\in D_4 \quad (x\in[1,e) \wedge y\in[0,1))$

$$F_{X,Y}(x,y)=\int_1^x\int_0^y\frac{2(1-v)}{u}\,dv\,du=2\int_1^x\frac{1}{u}\left(v-\frac{v^2}{2}\right)\Big|_0^y du=\int_1^x\frac{1}{u}(2y-y^2)\,du=(2y-y^2)\ln u\Big|_1^x=(2y-y^2)\ln x$$

- $(x,y)\in D_5 \quad (x\in[e,\infty) \wedge y\in[1,\infty))$

$$F_{X,Y}(x,y)=\int_1^e\int_0^1\frac{2}{x}(1-y)\,dy\,dx=\int_1^e\frac{2}{x}\left(y-\frac{y^2}{2}\right)\Big|_0^1 dx=\int_1^e\frac{1}{x}\,dx=\ln x\Big|_1^e=1$$